

Esercizio I: [4 pt]

Calcolare

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\arctan(x+y) - x - y}{x^2 + y^2}$$

$$\lim_{x,y \rightarrow 0,0} \frac{x+y - \frac{(x+y)^3}{3} - x - y}{x^2 + y^2}$$

$$= \frac{-(x+y)^3}{x^2 + y^2} = - \frac{x^3 + 3x^2y + 3xy^2 + y^3}{x^2 + y^2} = \frac{p^3(\dots)}{p^2} = 0$$

Esercizio II: (pt 8)

Calcolare

$$\iint_A \frac{x^2}{y} e^{xy} dx dy$$

dove  $A = \{(x,y) \in \mathbb{R}^2, \frac{1}{2x} \leq y \leq \frac{1}{x}, 2x^2 \leq y \leq 3x^2\}$ .

$$A = \{(x,y) \in \mathbb{R}^2$$

$$\frac{1}{2} < xy < 1 \vee 2 < \frac{y}{x^2} < 3\}$$

$$\begin{cases} xy = u \\ \frac{y}{x^2} = v \end{cases}$$

$$\begin{cases} x = \frac{u}{y} = \frac{u}{y} = \frac{u}{v^{1/3} u^{2/3}} = \frac{u^{1/3}}{v^{1/3}} \\ \frac{y}{(\frac{u}{y})^2} = v \end{cases}$$

$$y = (v u^2)^{1/3}$$

$$dx dy \rightarrow \det|J| du dv$$

$$y \cdot \frac{y^2}{u^2} = v$$

$$y^3 = v u^2 \quad y = (v u^2)^{1/3}$$

$$x = \frac{u^{1/3}}{v^{1/3}}$$

$$y = (v u^2)^{1/3}$$

$$\int \frac{u^{3/2}}{v^{2/3}} e^u$$

$$\int \frac{u^{3/2}}{v^{2/3}} (1 \cdot \cancel{u^{1/2}}) e^u$$

$$\frac{\left(\frac{u}{v}\right)^{2/3}}{v u^{1/3}} e^u \frac{1}{3v} du dv$$

$$\frac{u^{2/3}}{v^{2/3}} \frac{1}{v} e^u \frac{1}{3v}$$

$$\frac{1}{3v^2} e^u du dv$$

$$\det(J) : \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\sqrt{\frac{u}{v}}}{3u} \\ -\frac{\sqrt[3]{\frac{u}{v}}}{3v} \end{vmatrix}$$

$$x = \left(\frac{u}{v}\right)^{1/3} \begin{matrix} \nearrow \frac{\sqrt[3]{\frac{u}{v}}}{3u} \\ \searrow -\frac{\sqrt[3]{\frac{u}{v}}}{3v} \end{matrix} \quad \left(\frac{u}{v}\right)^{1/3} \quad \left(\frac{1}{v^{1/3}}\right) u^{1/3}$$

$$y = \sqrt[3]{v u^2} \nearrow \frac{\partial y}{\partial u} = \frac{1}{v^{1/3}} \frac{1}{3} u^{-2/3} \quad \frac{\partial x}{\partial u} = \frac{1}{3 v^{1/3} u^{2/3}}$$

$$\frac{\partial x}{\partial v} = u^{1/3} \left[ -\frac{1}{3 v^{4/3}} \right] = -\frac{u^{1/3}}{3 v^{4/3}}$$

$$\frac{1}{3v}$$

$$\frac{\partial y}{\partial u} =$$

$$\frac{1}{3} \int_2^3 \int_{\frac{1}{2}}^1 \frac{1}{v^2} e^u \, du \, dv \quad v^{-2}$$

$$\frac{e^{1/2}}{3} \int_2^3 \frac{1}{v^2} \, dv$$

$$\frac{e^{1/2}}{3} \left| -\frac{1}{v} \right|_2^3$$

$$e^{-\frac{1}{2}} \left| -\frac{1}{3} + \frac{1}{2} \right|$$

$$\frac{-2+3}{6}$$

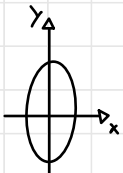
$$\frac{e^{-1/2}}{18}$$

Esercizio III: (pt 6)

Calcolare il lavoro del campo vettoriale  $F = \left( \frac{-y+x^3+xy^2}{x^2+y^2}, \frac{x+x^3+xy^2}{x^2+y^2} \right)$  lungo la curva descritta dall'equazione parametrica  $\gamma(t) = (2\cos(t), 3\sin(t))$ , con  $t \in [0, 2\pi]$ .

$$F = \left( \frac{-y+x(x^2+y^2)}{x^2+y^2}, \frac{x+x^3+xy^2}{x^2+y^2} \right) \rightarrow \dot{\gamma}(-2\sin t, 3\cos t)$$

$$F_1 = \left( -\frac{y}{x^2+y^2}; \frac{x}{x^2+y^2} \right)$$

$$F_2 = \left( x; x \right)$$


$$\frac{4p^2\cos^2\vartheta}{4} + \frac{9p^2\sin^2\vartheta}{9} = 1$$

$$\rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$$

$$\begin{cases} x = a p \cos\vartheta \\ y = b p \sin\vartheta \end{cases} \quad ab p d p d\vartheta$$

$$\text{rot}(F) = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2} - [-(x^2+y^2) + y(2y)]$$

$$= \cancel{x^2+y^2} - 2x^2 + \cancel{x^2+y^2} - 2y^2 = 0$$

$$\text{rot}(F) = 0$$

$$L(F, \gamma) = \int_0^{2\pi} F(\gamma(t)) \dot{\gamma}(t) dt$$

$$= \int_0^{2\pi} \left( -\frac{2\sin t}{4}; \frac{2\cos t}{4} \right) \left( -2\sin t; 2\cos t \right) dt$$

$$= \int_0^{2\pi} \left( -\frac{1}{2}\sin t; \frac{1}{2}\cos t \right) \left( -2\sin t; 2\cos t \right) dt$$

$$= \int_0^{2\pi} \sin^2\vartheta + \cos^2\vartheta$$

$$= \int_0^{2\pi} 1 d\vartheta = 2\pi$$

$$\cos(F_2) = 1$$

$$\int_0^1 \int_0^{2\pi} 6 p \, dp \, d\theta$$

$$6 \int_0^1 \int_0^{2\pi} p \, dp \, d\theta$$

$$\frac{3}{6} \cdot \frac{1}{2} \cdot 2\pi$$

$$6\pi$$

Esercizio IV: (pt 8)

Dato il campo vettoriale  $F = (\frac{x^2}{2}, \frac{y^2}{2}, z)$ , calcolarne il flusso attraverso la superficie laterale del paraboloide descritto dall'equazione  $z = 4 - (x^2 + y^2)$ ,  $z \in (0, 4)$  con il versore normale che punta verso l'alto.

$$\hookrightarrow x, y, 4 - (x^2 + y^2)$$

$$F = \iint F \cdot \vec{n} \, d\sigma$$

$$i \quad j \quad k$$

$$\det(J) =$$

$$\frac{\partial f}{\partial x} = 1, 0, 0$$

$$\frac{\partial f}{\partial y} =$$

$$\frac{\partial f}{\partial z}$$

## Discorso sul flusso

Flusso  $\rightarrow$  divergenza  $\rightarrow$  occhio perché si usa quando escono fuori numeri semplici

$$\iint F \vec{n} dS$$

Bisogna che  $n = \det(J) =$

$$\frac{\partial f}{\partial x}$$

$$\frac{\partial f}{\partial y}$$

$$\iint (F_x, F_y, F_z) (\det(J))$$

Con campo conservativo (aka  $\text{rot} = 0$ ) ho due strade percorribili:

① Potenziale

② oppure calcolo il lev. come  $L(F, \gamma) = \int F(\gamma) \dot{\gamma}(t) dt$

⚠ se la curva è chiusa  $L(F, \gamma) = 0$

⚠  $\gamma(t)$  la posso parametrizzare come un'altra curva in modo che abbia lo stesso punto d'inizio e lo stesso punto di arrivo

$$\text{Rot}(F) \neq 0; \quad \text{taccia} \quad L = \iint_E \text{rot}(F) dx dy$$